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Unraveling the effect of engagement and consistency in the results of the M6 forecasting competition

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ABSTRACT

The M6 competition evaluated investment performance over a period of one year, contributing to the efficient market hypothesis debate. This paper provides further insights into the outcomes of the competition by unraveling the effect that team engagement and performance consistency had on the final results. First, we identify three different types of engagement and investigate their relationship with portfolio efficiency, also making useful observations about the learning effect implied by a re-submission process. Then, we analyze the monthly performance of the teams and determine whether it aligned with their global performance or was affected significantly by extreme instances. Our results suggest that consistency is more important than engagement for making profitable investments. Nevertheless, we identify many cases where both regular portfolio updates and luck provided an advantage.

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1. Introduction

The M6 competition aimed to evaluate forecast accuracy in financial settings and assess the utility of forecasts in terms of investment performance . To do so, the competition considered a universe of 100 publicly traded assets (50 stocks and 50 ETFs) and employed a variant of the information ratio (IR), defined as the ratio of the portfolio return to the standard deviation of said return. Moreover, M6 included a total of 12 monthly submissions (every four weeks) with the objective to base its results on long-term cumulative returns and enhance the representativeness of its findings. The competition, whose design, execution, and results are summarized by Makridakis et al. (2023), attracted a total of 318 participants in 226 teams, 163 of which were included in the global leaderboard.[1]

According to the evaluation setup of the competition, teams were not obliged to update their submissions at every single round but could retain a previous submission for one or multiple future rounds. In other words, if for example a team made a single submission at the first month of the competition, this same submission would have been used to evaluate its performance across all submission points. Given that the competition covered a full calendar year, the purpose of this design attribute was two-fold: first, not to disqualify teams that may have unintentionally missed a submission point; and second, to support participants who did not wish to trade assets frequently but rather preferred to consider long-term investments. Ex post, this option led to a significantly low percentage of teams (about 16% of the 163 teams included in the global leaderboard) providing new submissions to all rounds, with almost one-quarter of the teams relying solely on their first submission. The organizers concluded that the overall correlation between the number of submissions made and the performance

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[1] Teams that entered the competition in the first month.

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of the team was minor, but observed that some winning teams did make multiple submissions. This observation suggests that active participation was probably a necessary but not sufficient condition for winning M6. Thus, it only scratches the surface of the importance of updating the investment decisions regularly or relying on previous ones for longer periods.

Another observation on the published results of the competition was that the investment performance of the teams often changed significantly across the submission rounds. That is, although some teams displayed a steady increase in portfolio returns, others seemed to base their final ranking on a few exceptional triumphs. Therefore, an interesting question emerges about the reasons that led each team to its final rank and, more importantly, about whether the evaluation setup of M6 managed to successfully distinguish consistently well-performing teams over more 'lucky' ones.

To that end, in this paper we discuss two questions:

- How crucial is the engagement of a team to win a forecasting competition like M6, and consequently, is it beneficial for investors to regularly update their investment decisions?
- How decisive was the consistent performance of a team to its distinction over participants who sporadically performed exceptionally?

In practice, the first question is more relevant for determining whether optimal results typically rely on consistently readjusted portfolios or on a general understanding of the market with minimal interference. Equivalently, it is also useful for providing evidence about whether trading can yield higher profits than long-term investments (Arnott & Lovell, 1993). The second question, although more relevant for assessing the overall robustness of the setup used in the competition for determining the winners, also has implications for investors, questioning whether an investment strategy should primarily aim at yielding consistent small revenues over time or whether it should attempt to exploit volatile periods and high-risk portfolios to achieve higher returns (Markowitz, 1952). The following two sections investigate the aforementioned questions, while the last one concludes the paper.

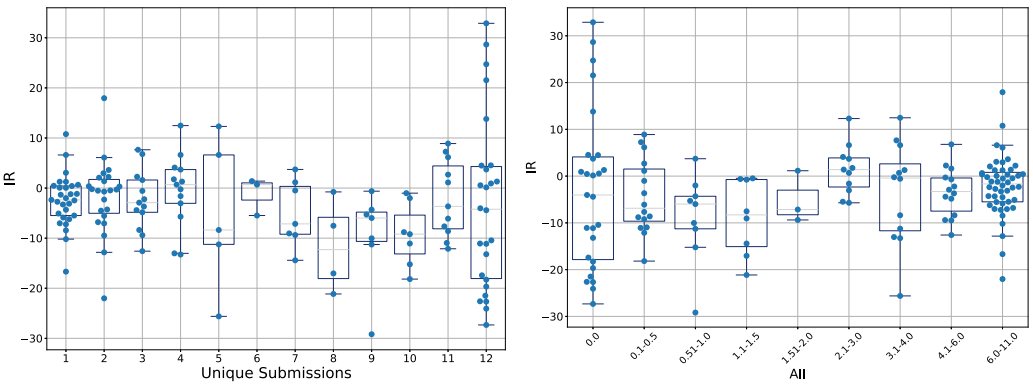
2. Does engagement matter?

We first turn our focus on the engagement of the participating teams in the competition with the objective of investigating its effect on investment performance. In theory, teams that updated their submissions often should have had an advantage over those that did so less frequently or even retained their initial submissions throughout the competition. This is because frequent updating allows the methods used for constructing the portfolios to exploit new information and to respond quickly to possible changes in the market. Nevertheless, such updates are often irrelevant in practice, with intermediate updating strategies resulting in similar or even better outcomes (Spiliotis & Petropoulos, 2024). This can be particularly relevant in financial forecasting settings where the signal-to-noise ratio is low and, therefore, limited insights can be extracted from the new information that becomes available. Effectively, this analysis can be deemed a simplified comparison of performance between traders and investors, where the former adjust their portfolios repeatedly using technical analysis tools, while the latter make long-term investments by relying on company, industry, and economic fundamentals.

2.1. Criteria for measuring engagement

Before proceeding with the analysis, it is important to define 'engagement', since the term can be interpreted in various ways. In this regard, we identify three types of engagement, based on the following criteria:

- C1 – Frequency of submissions: Teams that have made more submissions overall are considered to be more engaged (Sujit R. Das & Goyal, 2014). Based on this criterion, a team that made eight submissions was more engaged than a team that made six submissions.
- C2 – Idle interval of submissions: Teams that, on average, made submissions within shorter intervals are considered to be more engaged. Based on this criterion, a team that made four consecutive submissions at the beginning of the competition was less engaged than a team that made four submissions, each at the beginning of each quarter.
  To quantify idleness, we introduce the average idle interval (AII) measure, defined as follows:
  $$\mathrm{AII} = \sqrt{\frac{\sum_{i=1}^{n} t_i^2}{n}},$$
  where t is the number of periods (in this case, months) that have passed without making a new submission, being zero if the previous submission is revised in the following period, one if it is revised after one period, and so on; and n is the total number of submissions made. Observe that by squaring t, the measure penalizes longer idle intervals more, thus diversifying teams, even if they made the same number of submissions. In that sense, although AII looks similar to the inter-demand interval introduced by Syntetos and Boylan (2005) for quantifying intermittency of demand, it provides different insights.
- C3 – Differentiation of submissions: Teams that changed their investment strategies before making new submissions are considered to be more engaged. Based on this criterion, a team that made four submissions using the same investment strategy was less engaged than a team that made two submissions using a different strategy each time.
  The reason for introducing C3 is that, even between similarly active teams, the level of engagement may still vary. Let us assume that a team developed an investment strategy at the very beginning of the competition and reapplied it unmodified in every submission period based on new information that



(a) Frequency of submissions (C1).

(b) Average idle interval of submissions (C2).

(c) Number of strategy changes (C3).

Fig. 1. Relationship between the IR and the engagement criteria considered.

became available. Undoubtedly the team was engaged, but made no additional effort to improve the performance of its initial approach. As such, teams that tried to adjust their investment strategies based on the experience acquired and feedback received from the leaderboard across the competition should be regarded to be more engaged. Since there is, unfortunately, no accurate way to track the changes made in the investment strategy of each team or monitor the actual workload required for making said changes, we base our analysis on the same process used by the organizers of the competition for measuring strategy consistency. That is, we consider the changes made in terms of exposure, diversification, investment weight range, and investment directionality in the investment strategies of the teams (see Hypothesis No. 5 in Makridakis et al., 2023).

Note that the three means of classification utilized do not account for teams that were active but decided not to regularly make new submissions. This is because, based on the information available, it is impossible to discern whether a team chose as part of their strategy to skip a submission, or just missed it. Consequently, we aggregate the performance of teams that intentionally did not update their submissions with those that forgot or did not have time to do so—a practice that may distort the results. Having said that, we clarify that we also attempted to perform a more detailed analysis on the performance of teams whose engagement could indirectly be confirmed. A way to identify such teams is to examine, for instance, whether the forecasts they submitted were always identical to those of the benchmark,[2] in the sense that if this was true, then these teams were solely focused on the investment part of M6. Unfortunately, this condition applies to a small group of only 12 teams that ranked from 26th to 134th, rendering the extraction of any concrete conclusion impossible.

2.2. Analysis and initial findings

To conduct our analysis, we consider only the 148 teams included in the 'Global' leaderboard whose investment submissions were not identical to the benchmark, excluding those that did not submit original investment decisions, and drawing conclusions based on the overall performance of the teams across all 12 submission points of the competition.

Starting with C1, we analyze the association between the IR and the frequency of submissions. Fig. 1(a) illustrates this relationship by presenting the distribution of

[2] The benchmark took equal long positions for all 100 assets (DeMiguel et al., 2009).

the IR for groups of teams with the same number of submissions. We observe that the median value of most groups is lower than zero, indicating that making multiple submissions did not guarantee better outcomes. Moving to C2, Fig. 1(b) illustrates the IR of the participating teams versus the AII. In line with C1, there is no clear relationship between idleness and investment performance, since the IR of the teams with an AII value of zero largely overlaps with that of the rest of the teams. Fig. 1(c) showcases the connection between the IR of the teams and the number of times they altered their investment strategy according to C3. We find that the vast majority of teams retained consistent strategies throughout the competition, making no or only one strategy change, while considerably fewer teams differentiated their original submission more than twice. Due to this sample imbalance and the high variability of IR in each group, no definitive conclusions can be reached about whether maintaining a consistent investment strategy had a positive or negative effect on performance. To formally and jointly investigate the above relationships, we also estimated the following regression model that connects the rank of the teams with the examined engagement criteria. None of the coefficients were found to be statistically significant.

$$\text{Rank} = a + b \times \text{C1} + c \times \text{C2} + d \times \text{C3} \quad (1)$$

With our results being inconclusive, we proceed by analyzing the effect of engagement from a different angle, investigating whether an 'early stopping' strategy in the re-submission process would have improved the IR of the teams. Specifically, for each team, we calculate the IR at the end of the competition, assuming that they would have stopped making new submissions at an earlier point than they actually did. For example, for teams that made two submissions throughout the competition, we calculate their performance assuming they would have submitted only their first submission instead of their first and second submissions. Similarly, for teams that made submissions at all 12 points, we calculate an alternate IR 11 times, one for each point they could have stopped their submissions. Fig. 2 presents the average gain in investment performance for the groups of teams that made a total of three, six, nine, and 12 submissions. There is a clear trend for all four groups,[3] indicating that as the teams made additional submissions, their IR scores dropped on average. This observation suggests that re-submitting added little (if not negative) value on average.

2.3. Differentiation of the top performers

Given that the previous analysis involved all the participating teams, even though they may have had very different portfolio construction skills, we now turn our attention to the top-performing teams of the competition, aiming to identify patterns among participants of higher potential. To that end, we begin by re-estimating the regression model of Eq. (1) for the top 25% of teams. As shown in Table 1, the frequency of submissions (C1) does report a statistically significant positive effect over the rank of the teams. Therefore, there is some evidence that frequent portfolio updates could act beneficially, provided that relatively effective approaches are used to construct the portfolios (Mark Kritzman & Turkington, 2010).

Table 1
Coefficient significance of a linear regression model relating the rank of the top 25% of teams in terms of IR with measures suggesting engagement (C1, C2, and C3). A variable is generally considered significant when $p < 0.05$.

Variable	Coefficient	p-value
Intercept	28.48	0.002
C1	−2.06	0.030
C2	−0.80	0.445
C3	2.77	0.086

The previous finding becomes more evident if we focus on the top 15 performing teams of the competition. As shown in Fig. 3(a), the average top performer made more than seven submissions, five of the top six performing teams made 12 submissions, and only one team made a single submission. Moreover, in contrast to the results in Fig. 2 that focus on the complete population of the teams, Fig. 3(b) suggests that all top-performing teams were benefited highly by updating their previous submissions.[4] Specifically, by implementing the 'early stopping' strategy explained above for the top 15 performing teams, we find that resubmitting would have a positive effect overall, with the impact being statistically significant in 30% of the cases. These observations largely confirm our previous conclusion, suggesting a positive relationship between engagement and investment performance, as long as effective methods are utilized for constructing the portfolios.

2.4. Engagement as a learning process

In previous forecasting competitions, feedback was found to significantly improve performance over benchmarks (Athanasopoulos & Hyndman, 2011). Therefore, we move forward by evaluating the learning effect that submission updating implies, as reflected in the percentage of teams that managed to outperform the market over the competition. For the sake of simplicity, we focus on the two extremes, i.e., the teams that made a single versus 12 submissions overall. Fig. 4 visualizes the results, suggesting that the number of teams that beat the benchmark varied significantly across the months, mostly depending on whether the market was bullish or bearish (outperforming the market becomes more challenging when its performance increases, and vice versa). In that sense, since in the second semester market returns almost doubled compared to the first semester, one would expect that the percentage of teams performing better than the

[3] As an example, a linear trend model fitted to the 12 submissions line of Fig. 2 has a correlation coefficient $r = 0.62$ and a p-value (on the coefficient responsible for capturing trend) of 0.044.

[4] As an example, a linear trend model fitted to the winner's line of Fig. 3(b) has a correlation coefficient $r = 0.75$ and a p-value of 0.005. Also, the correlation coefficient was positive for all examined teams, although not always statistically significant.

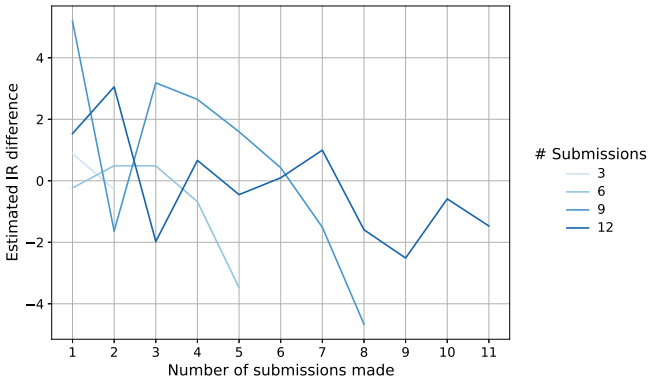
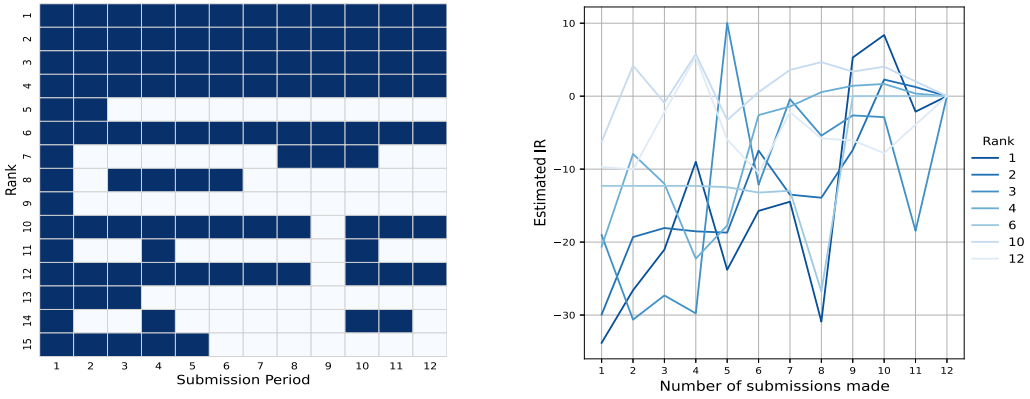


Fig. 2. Average estimated IR difference of teams assuming they would have stopped making new submissions at an earlier point than they actually did in M6. Positive numbers suggest better performance than originally reported.



(a) Submissions made by the top 15 performing teams in M6. Dark cells indicate new submissions, while white cells no submissions.

(b) Estimated IR difference of selected top performing teams assuming an "early stopping" strategy.

Fig. 3. Analysis of the top 15 performing teams.

benchmark would drop on average. That was indeed the case for teams that made a single submission throughout the competition (their percentage dropped on average from 42% to 39%), but not for the teams that submitted on all submission periods (their percentage increased on average from 44% to 49%). This variation in the behavior of the two groups could indicate that teams had to get familiar with the setup of the competition and experiment with their investment strategies for some time before being able to capitalize on the experience and insights gained. To account for this learning curve, the organizers introduced a trial run just before the competition started. However, due to the limited participation of teams in said run and its short duration, its effect might have been insignificant in practice. As such, it is possible that the advantages of engagement would have become more evident if the competition had lasted longer than 12 months.

3. Consistent performance or big wins?

Our second research question revolves around the factors that allowed teams to secure top positions in the investment track of the competition, and more specifically on whether a consistently good performance of a team was sufficient to distinguish it over teams that reported great performance in certain periods. In principle, in order for a portfolio to be efficient in the long run, it has to perform well consistently instead of relying on sporadic big wins. In this regard, given a robust competition setup, such approaches should have been awarded more.

Before proceeding with the analysis, we deem as 'consistent' a team that had stable returns to risk ratios throughout the competition, i.e., reported monthly IR values that did not vary significantly around the corresponding global IR. In this direction, Fig. 5 presents the IR of the teams versus their average monthly IR. As seen, the correlation between the two measures is particularly strong, suggesting that teams that performed well overall were also those that performed well in most of the individual evaluation periods. Indeed, by observing the quantile regression lines of Fig. 5, distinguishing the upper and lower tails of the distribution, we identify only five teams that performed slightly worse than expected based on their average monthly performance (teams above the 95% line), as well as a total of just four teams that performed slightly better than their average

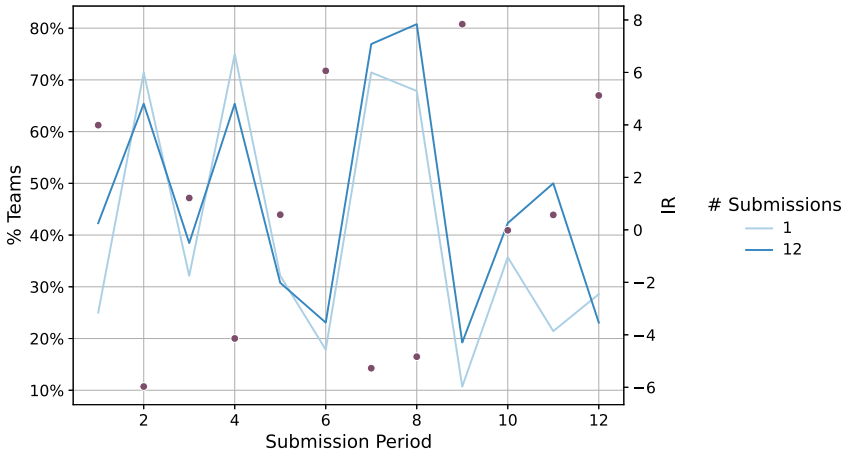


Fig. 4. Percentage of teams beating the benchmark in each submission period. Results are presented only for teams that made one or 12 submissions overall. The dots indicate the IR (secondary axis) of the benchmark at every period.

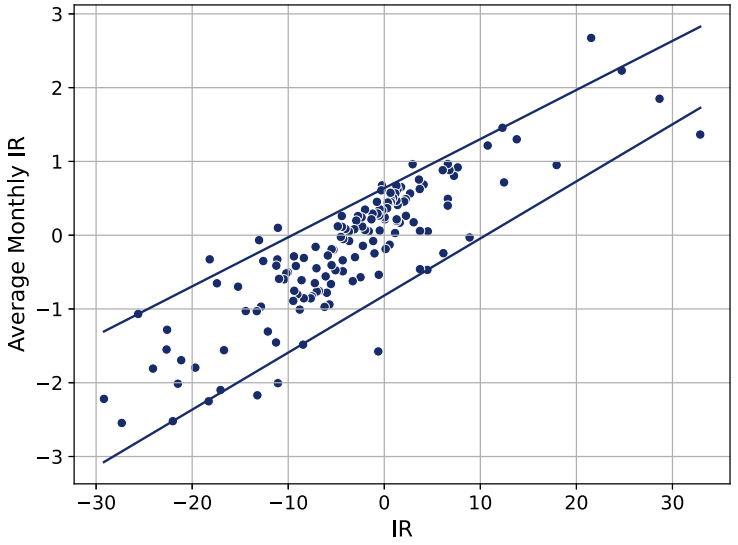


Fig. 5. Relationship between the IR and the average monthly IR ($r = 0.86$, $p < 0.001$). The quantile lines at 5% and 95% are also provided.

monthly performance would imply (teams below the 5% line).

Considering that teams with a less volatile IR should have been rewarded with higher ranks, Fig. 6 presents the relationship between the teams' standard deviation of monthly IR values and their respective rank. In order to cancel out randomness and facilitate the identification of any correlation, the results are aggregated in groups of 15 teams. There is a slight trend observed in the data, confirming that, overall, consistent teams managed to perform better than the teams that exhibited larger fluctuations in terms of the IR.

Given that variations in the IR should mainly be identified in portfolios that include more volatile assets, Fig. 7(a) illustrates the relationship between the overall risk taken by the teams and their final rank, using the denominator of the IR as a measure of risk. Similarly, Fig. 7(b) shows the relationship between the percentage of capital the teams invested in stocks (in general, ETFs are credited to be less risky than stocks) and their respective rank. In both cases we find that more risky strategies resulted in higher ranks, with conservative teams achieving lower ranks in the competition. Nevertheless, there is an evident exception to the underlying trend, as the top 15 performing teams managed to achieve the highest ranks with the lowest average risk, i.e., construct particularly efficient portfolios. To validate the relationship between risk and ranks more formally, we also performed a pairwise one-way ANOVA between three groups of risk (low, medium, and high) to test whether there is a significant difference between their mean rank or IR. While testing multiple different risk ranges to create the three groups, we did not manage to identify any statistically significant differences between the means of any pair.

Moving forward and focusing on the top-performing teams, Fig. 5 suggests that the winning team scored a global IR somewhat higher than expected, whereas the team ranked 4th marginally underperformed based on its

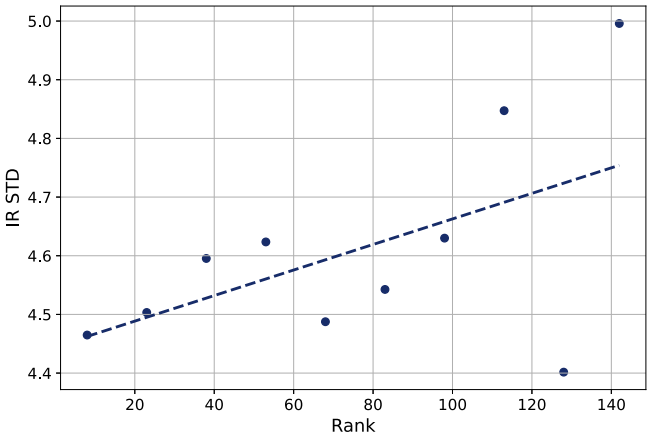
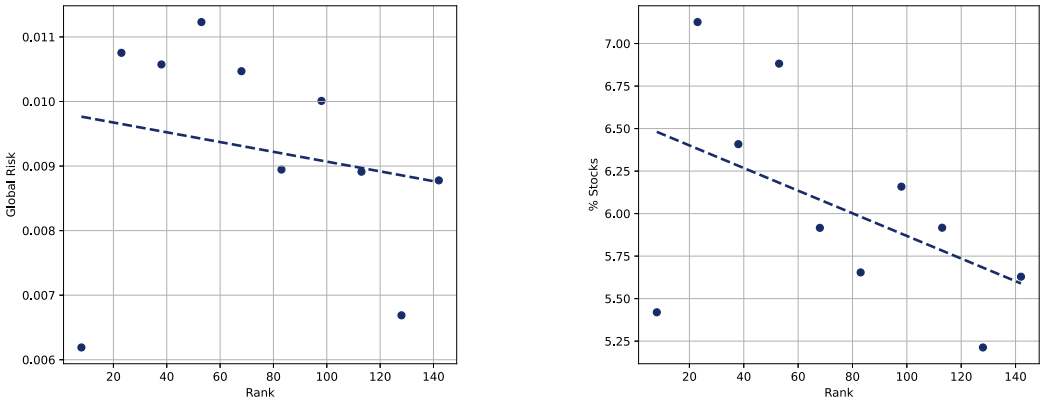


Fig. 6. Relationship between the standard deviation of monthly IR values and rank ($r = 0.54$, $p = 0.106$). Results are grouped in buckets of 15.



(a) Relationship between the risk and the rank of the teams (r=0.2, p=0.579).

(b) Relationship between the percentage of capital allocated in stocks and the rank of the teams (r=0.49, p=0.155).

Fig. 7. Risk taken by the teams and its effect on their rank. Results are grouped in buckets of 15.

average monthly IR. To investigate this premise further, in Fig. 8 we present the cumulative returns of the top 10 performing teams along with those of the benchmark and the IVV ETF, since it mimics the performance of the S&P 500 Index. It is apparent that, compared to the benchmark and the IVV ETF, all teams managed to maintain a better balance between the risk taken and the returns realized. Nevertheless, some teams opted for much riskier strategies to increase their chances of securing a higher position in the global leaderboard. Specifically, we observe that the winning team followed a relatively riskier approach, delivering two big wins ($>$ 10% monthly returns) during the second and ninth months of the competition, successfully enabling its differentiation. On the contrary, the team ranked 4th chose a lower-risk investment approach with moderate exposure in the beginning of the competition and minimal exposure during its second half, which led to lower returns and, hence, an inferior IR.

Fig. 9 further supports the analysis conducted above, illustrating the distribution of the monthly IR of each of the 10 highest ranked teams. We observe that the first nine teams all reported at least an exceptional IR (above the 90th quantile line) and practically avoided lousy scores (below the 10th quantile line). At the same time, however, the median performance of the teams is not perfectly aligned with their global positions, since the median of the 1st team is well below the rest of the medians, while the medians of the 4th and 8th teams surpass them.

Summarizing our previous observations, we can conclude that, in general, teams that performed consistently well across the 12 months of the competition were also those that ranked at the top of the global leaderboard. In this regard, the setup of the M6 competition can be regarded as successful. Nevertheless, given the challenging nature of the competition and market uncertainty, it seems that teams ultimately had to take some reasonable risk to distinguish themselves from the rest of the participants, hoping for some relatively big wins that would effectively counterbalance previous, small losses.

4. Conclusion

The M6 competition provided a rigorous test bed for evaluating the efficient market hypothesis along with

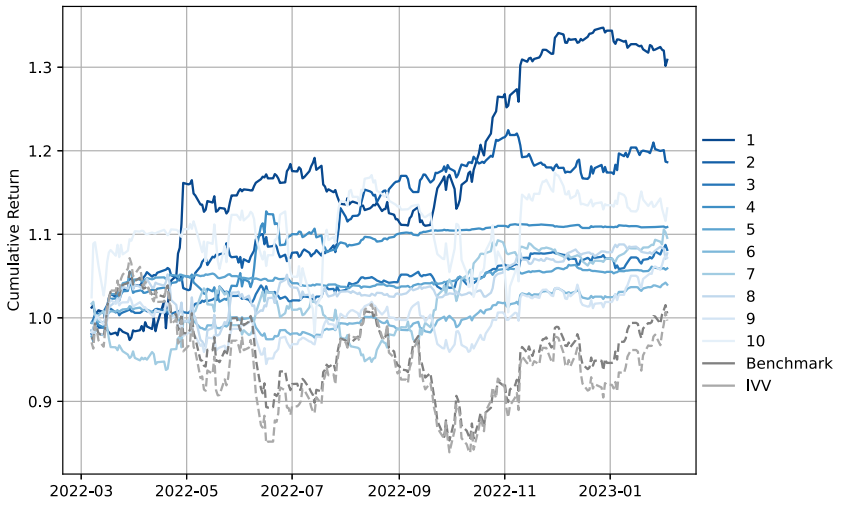


Fig. 8. Cumulative returns of the top 10 performing teams, the benchmark, and the IVV ETF.

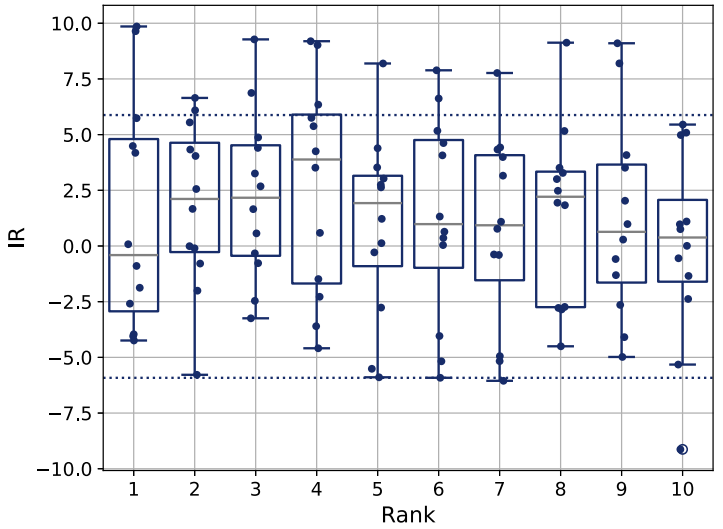


Fig. 9. Distribution of the IR at a monthly level for the top 10 performing teams of the competition. The two parallel lines represent the 10% and 90% quantiles of the monthly IR distribution across all 148 teams.

nine more hypotheses introduced by the organizers before its launch. More importantly, it measured the benefits of accurate forecasting and assessed the importance of forecasting when making investment decisions. Inspired by the findings and challenges of M6, in this discussion paper we investigated the effect of team engagement and performance consistency on the results and global ranking of the competition, assuming that both should have had a positive impact. Note that having no available data indicating the rationale behind each team's portfolio construction process limited the revelation of any causal relationships between those factors. Nevertheless, we believe that the key insights provided by our analysis shed light on how each factor influenced the achievement of higher ranks in the competition.

In terms of engagement, our analysis indicated that neither the frequency of submissions nor the intervals between consecutive submissions significantly influenced the investment performance of the teams. Moreover, the results suggested that even when the teams made additional efforts in revising their investment strategies, no improvements were guaranteed in terms of the IR. This lack of a relationship between the engagement of the teams and the performance of their investments can, however, be attributed to the effectiveness of the approaches utilized by each team for constructing the portfolios, which varied significantly, rendering global comparisons unsuccessful. Indeed, when focused on the top-performing teams, where the approaches used for portfolio construction were similarly efficient, we found that the

vast majority of the teams considered frequent updates, being active across the whole duration of the competition. More importantly, we found that the additional submissions made by these teams helped them improve their performance on average, in contrast to the majority of the other teams, where new submissions tended to deteriorate performance.

By analyzing the consistency of investment performance, we found that teams that maintained stable performance across the 12 months of the competition typically secured top positions on the global leaderboard as well. However, our results also suggest that many of the winners aimed for substantial returns at some points of the competition to offset earlier setbacks, taking calculated risks that allowed them to stand out from their peers. In fact, teams that took higher risks were ranked higher in the leaderboard on average. The fact that we did not manage to identify any strong relationship between portfolio risk and team ranks further supports the above insight. Consequently, we conclude that although the setup of M6 provided a fair testing ground—awarding teams that properly balanced risk with expected returns—luck was also relevant to some extent. By extrapolating this finding, there is evidence that traders and investors have similar opportunities for making profitable investments.

Another interesting finding of our investigation was that teams that made several submissions seemed to have gradually improved their performance over the benchmark compared to those that did not update their initial submissions, thus gaining some advantage over time. Nevertheless, these improvements were marginal on average and observable after the first half of the competition. To that end, we conclude that providing the participants of forecasting competitions with a sufficient number of trial runs to get familiar with the objectives and particularities of the task at hand could be helpful for revealing their full potential, at least in cases where teams experiment with relatively efficient investment strategies. Yet, the design choices of any competition should always trade off with practical concerns like costs and the effort required to participate actively in them (Makridakis et al., 2022).

Forecasting competitions employing live evaluations on real data across multiple periods definitely allow for more robust and trustworthy assessments, enabling teams to test different methods and revise their approaches based on the feedback received. Although, in M6, market uncertainty naturally narrowed the beneficial effect of engagement, we believe that future competitions should continue encouraging this practice, especially when the utility of forecasts is being evaluated in addition to their accuracy.

CRediT authorship contribution statement

Anastasios Kaltsounis: Writing – original draft, Visualization, Methodology, Data curation, Conceptualization. **Evangelos Theodorou:** Writing – original draft, Validation, Conceptualization. **Evangelos Spiliotis:** Writing – review & editing, Conceptualization. **Vassilios Assimakopoulos:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data and code are gathered as part of a reproducibility package and is readily available at: https://github.com/TasosKaltsounis/Unraveling-the-effect-of-engagement-and-consistency-in-the-results-of-the-M6-forecasting-competition.

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